Ellipses

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Standard form**

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:

a the vertices $(\pm 5, 0)$

b the foci $(\pm 4, 0)$

c the endpoints of the minor axis $(0, \pm 3)$

d the eccentricity $e = \frac{4}{5}$

(a) $(\pm 5, 0)$; (b) $(\pm 4, 0)$; (c) $(0, \pm 3)$; (d) $e = \frac{4}{5}$

2 **[Foundational]** • **Target: Standard form**

Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.

Major axis vertical with $a = 5$, $c = 3$, so $b^2 = 25 - 9 = 16$: $\frac{x^2}{16} + \frac{y^2}{25} = 1$.

3 **[Foundational]** • **Target: Standard form**

Consider the ellipse $9x^2 + 4y^2 = 36$.

a Write the equation in standard form. $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

b State the lengths of the major and minor axes. Major axis $2a = 6$ (vertical); minor axis $2b = 4$.

c Find the coordinates of the foci. $c^2 = 9 - 4 = 5$, so foci $(0, \pm \sqrt{5})$.

(a) $\frac{x^2}{4} + \frac{y^2}{9} = 1$.; (b) Major axis $2a = 6$ (vertical); minor axis $2b = 4$.; (c) $c^2 = 9 - 4 = 5$, so foci $(0, \pm \sqrt{5})$.

4 **[Foundational]** • **Target: Applications**

A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?

$c = \sqrt{400 - 144} = \sqrt{256} = 16$, so the foci are $2c = 32.0$ m apart.

5 **[Foundational]** • **Target: Applications**

An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.

$a = 6$: $\frac{9}{36} + \frac{12}{b^2} = 1$ gives $\frac{12}{b^2} = \frac{3}{4}$, so $b^2 = 16$: $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

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6 [Foundational] • Target: Ellipses centered away from the origin

In an analytical model, for the ellipse $\frac{(x-2)^2}{16}+\frac{(y+1)^2}{9}=1$, find:

a the center $(2,\,-1)$

b the vertices $(2\pm 4,\,-1)=(6,\,-1),(-2,\,-1)$

c the foci $c=\sqrt{16-9}=\sqrt{7}$: $(2\pm\sqrt{7},\,-1)$

(a) $(2,\,-1)$; (b) $(2\pm 4,\,-1)=(6,\,-1),(-2,\,-1)$; (c) $c=\sqrt{16-9}=\sqrt{7}$: $(2\pm\sqrt{7},\,-1)$

7 [Foundational] • Target: Ellipses centered away from the origin

Complete the square to write $4x^2+y^2-8x+6y+9=0$ in standard form, and state the center.

$4(x^2-2x)+(y^2+6y)=-9$: $4(x-1)^2-4+(y+3)^2-9=-9$, so $4(x-1)^2+(y+3)^2=4$, i.e. $(x-1)^2+\frac{(y+3)^2}{4}=1$. Center: $(1,\,-3)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Eccentricity and orbital applications

Compute the eccentricity of each ellipse, and state which is more circular (closer to $e=0$):

a $\frac{x^2}{100}+\frac{y^2}{36}=1$ $c=\sqrt{100-36}=8$: $e=\frac{8}{10}=0.8$

b $\frac{x^2}{25}+\frac{y^2}{24}=1$ $c=\sqrt{25-24}=1$: $e=\frac{1}{5}=0.2$ — this one is more circular

(a) $c=\sqrt{100-36}=8$: $e=\frac{8}{10}=0.8$; (b) $c=\sqrt{25-24}=1$: $e=\frac{1}{5}=0.2$ — this one is more circular

9 [Intermediate] • Target: Eccentricity and orbital applications

A planet's elliptical orbit has semi-major axis $a=150$ million km and eccentricity $e=0.02$. Find the distance from the center to a focus, c, and hence the closest approach distance to the sun ($a-c$), to two decimal places (in millions of km).

$c=ea=0.02(150)=3$ million km. Closest approach $=a-c=150-3=147.00$ million km.

10 [Intermediate] • Target: Graphing an ellipse

The upper half of the ellipse $\frac{x^2}{16}+\frac{y^2}{4}=1$ is shown (as $y=2\sqrt{1-x^2/16}$). Use it to estimate the y-value at $x=2$, then verify algebraically.

From the graph, $y\approx 1.7$ at $x=2$. Algebraically: $y=2\sqrt{1-4/16}=2\sqrt{0.75}\approx 1.73$ ✓.

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11 [Intermediate] • Target: The ellipse's defining property

State the geometric (sum-of-distances) definition of an ellipse, and use it to verify that the point $(0, 3)$ lies on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with foci $(\pm\sqrt{7}, 0)$.

An ellipse is the set of points where the sum of distances to the two foci is constant ($= 2a$). At $(0, 3)$: distance to each focus $= \sqrt{7 + 9} = 4$, so the sum is $8 = 2a = 2(4)$ ✓, confirming the point lies on the ellipse.

12 [Intermediate] • Target: Writing equations from a description

Write the standard-form equation of an ellipse centered at $(3, 2)$ with a horizontal major axis of length 10 and minor axis of length 6.

$a = 5$, $b = 3$: $\frac{(x-3)^2}{25} + \frac{(y-2)^2}{9} = 1$.

13 [Intermediate] • Target: Writing equations from a description

Write the standard-form equation of an ellipse with vertical major axis, center $(-1, 4)$, $a = 7$, and $b = 2$.

$\frac{(x+1)^2}{4} + \frac{(y-4)^2}{49} = 1$.

14 [Intermediate] • Target: Systems with ellipses

Find the points of intersection of $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the line $y = x$.

Substitute $y = x$: $\frac{x^2}{9} + \frac{x^2}{4} = 1 \Rightarrow \frac{4x^2 + 9x^2}{36} = 1 \Rightarrow 13x^2 = 36 \Rightarrow x = \pm\frac{6}{\sqrt{13}}$. Points: $\left(\pm\frac{6}{\sqrt{13}}, \pm\frac{6}{\sqrt{13}}\right)$ (same sign pairs).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 9.02 Geometric Sequences and Series

State the formula for the area of an ellipse with semi-axes a and b, and use it to find the area of $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

Area $= \pi ab$. Here $a = 5$, $b = 4$: Area $= 20\pi \approx 62.83$ square units.

16 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 9.03 The Binomial Theorem

An elliptical track for a running field has a semi-major axis of 60 m and semi-minor axis of 40 m. Find the exact area enclosed by the track, and give a decimal approximation to the nearest square metre.

Area $= \pi(60)(40) = 2400\pi \approx 7539.8$ m^2.

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17 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 1.01 Functions and Function Notation

Two ellipses both have eccentricity $e = 0.6$: one has $a = 10$, the other has $a = 20$. Find b for each, and explain why they are the same shape but different sizes (similar ellipses).

For $a = 10$: $c = ea = 6$, $b^2 = 100 - 36 = 64 \Rightarrow b = 8$. For $a = 20$: $c = 12$, $b^2 = 400 - 144 = 256 \Rightarrow b = 16$. Since $\frac{b}{a} = \frac{8}{10} = \frac{16}{20} = 0.8$ is the same ratio for both, the ellipses are geometrically similar (same shape, scaled by a factor of 2).

18 [Challenging] • Target: Systems with ellipses • Cross-Lesson: 1.02 Domain and Range

Find the standard equation of the ellipse centered at the origin with horizontal semi-major axis $a = 5$ and semi-minor axis $b = 3$.

The standard equation of an ellipse centered at $(0, 0)$ is: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ Substitute $a = 5$ and $b = 3$: $\frac{x^2}{5^2} + \frac{y^2}{3^2} = 1 \Rightarrow \mathbf{\frac{x^2}{25} + \frac{y^2}{9} = 1}$

19 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 1.03 Average Rate of Change

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor: $\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56 (3x)^3 = 27x^3 (-2y^2)^5 = -32y^{10}$ Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32) x^3 y^{10} = 56 \times (-864) x^3 y^{10} = -48384 x^3 y^{10}$.

20 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Standard form

The standard equation of an ellipse centered at origin with semi-axes $a = 4, b = 3$ along x, y respectively is:

a. $\frac{x^2}{4} + \frac{y^2}{3} = 1$

b. $x^2 + y^2 = 7$

c. $\frac{x^2}{16} + \frac{y^2}{9} = 1$ ✓

d. $\frac{x^2}{9} + \frac{y^2}{16} = 1$

Applying Conic Sections principles, Standard ellipse. Therefore, the verified correct option is $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

22 [Foundational] • Target: Applications

An ellipse $\mathcal{E}$ and a hyperbola $\mathcal{H}$ are confocal, sharing the foci $F_1(-4, 0)$ and $F_2(4, 0)$. The eccentricity of $\mathcal{E}$ is $\frac{4}{5}$, and the eccentricity of $\mathcal{H}$ is 2. If P is an intersection point of $\mathcal{E}$ and $\mathcal{H}$ in the first quadrant, what is the area of ΔPF_1F_2?

a. $6\sqrt{3}$ ✓

b. $12\sqrt{3}$

c. $3\sqrt{3}$

d. $4\sqrt{5}$

We can solve this using either the geometric focal definitions or algebraic coordinate equations. **Method 1: Focal Distance Properties** The foci are located at $(\pm c, 0)$ where $c = 4$, so the base of ΔPF_1F_2 along the x-axis has length $|F_1F_2| = 2c = 8$. 1. For ellipse $\mathcal{E}$ with eccentricity $e_E = \frac{c}{a_E} = \frac{4}{5}$: $a_E = \frac{c}{e_E} = \frac{4}{4/5} = 5$ By the focal definition of an ellipse, any point P on $\mathcal{E}$ satisfies: $|PF_1| + |PF_2| = 2a_E = 10$ 2. For hy.

23 [Foundational] • Target: Ellipses centered away from the origin

For the ellipse $b^2x^2 + a^2y^2 = a^2b^2$, confirm this equals standard form.

a. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ✓

b. $x^2 + y^2 = a^2 + b^2$

c. $ax + by = ab$

d. $b^2 + a^2 = 1$

Applying Conic Sections principles, Divide both sides by a^2b^2. Therefore, the verified correct option is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

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24 [Foundational] • Target: Eccentricity and orbital applications

The standard form of a circle centered at (h, k) with radius r is:

a. $x^2/r + y^2/r = 1$

b. $(x + h)^2 + (y + k)^2 = r^2$

c. $(x - h)^2 + (y - k)^2 = r^2$ ✓

d. $x^2 + y^2 = r$

Applying Conic Sections principles, Definition of circle. Therefore, the verified correct option is $(x - h)^2 + (y - k)^2 = r^2$.

25 [Foundational] • Target: Graphing an ellipse

Write the standard form of the parabola with vertex at origin opening upward with $p = 3$.

a. $x^2 = 12y$ ✓

b. $y^2 = 3x$

c. $x^2 = 3y$

d. $y^2 = 12x$

Applying Conic Sections principles, $x^2 = 4py = 12y$. Therefore, the verified correct option is $x^2 = 12y$.

26 [Foundational] • Target: The ellipse's defining property

Find the polar form of the conic with focus at origin, directrix $x = d > 0$, eccentricity e.

a. $r = \frac{d}{1 - e\cos\theta}$

b. $r = \frac{e}{1 + d\cos\theta}$

c. $r = \frac{d}{e\cos\theta}$

d. $r = \frac{ed}{1 + e\cos\theta}$ ✓

Standard polar conic with focus at origin, directrix to the right.

27 [Foundational] • Target: Writing equations from a description

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?

a. 12 ✓

b. 8

c. 4

d. -3

We are given the quadratic function $f(x) = x^2 - 6x + 5$ and a horizontal line $y = k$ that intersects the parabola at two points separated by a horizontal distance of 8. **Method 1: Using the axis of symmetry** The axis of symmetry for a parabola in standard form $y = ax^2 + bx + c$ is given by: $x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$ Because parabolas are symmetric about their axis of symmetry, the two intersection points on the horizontal line $y = k$ must lie equidistant from $x = 3$. Since the total dist.

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28 [Foundational] • Target: Systems with ellipses

Let m be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots z_1 and z_2. In the complex plane, the origin $O(0)$ and the points representing z_1 and z_2 form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

a. $15\sqrt{3}$ ✓

b. $12\sqrt{3}$

c. $\frac{15\sqrt{3}}{2}$

d. $9\sqrt{3}$

We solve the problem by synthesizing Vieta's formulas with the geometric properties of complex numbers. **Step 1: Condition for an Equilateral Triangle with the Origin** In the complex plane, three points 0, z_1, and z_2 form the vertices of an equilateral triangle if and only if one non-zero vertex is obtained by rotating the other about the origin by $\pm\frac{\pi}{3}$: $\frac{z_2}{z_1} = e^{\pm i\pi/3} = \frac{1 \pm i\sqrt{3}}{2}$ This is equivalent to the quadratic in $\frac{z_2}{z_1}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Area of an ellipse

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function
$f(x) = \begin{cases} \frac{3}{8}x^2, , 0 \le x \le 2, ; 0, , \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$ ✓

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

Let A denote the area of the rectangle and $L_{\min}$ denote the length of the shorter side. 1. **Determine the condition for $A \ge \frac{3}{4}$:** The area is given by $A = X(2 - X) = 2X - X^2$.
$2X - X^2 \ge \frac{3}{4} \Leftrightarrow X^2 - 2X + \frac{3}{4} \le 0 \Leftrightarrow (X - \frac{1}{2})(X - \frac{3}{2}) \le 0$ Thus, $A \ge \frac{3}{4}$ corresponds to the interval $E_1 = \left[\frac{1}{2}, \frac{3}{2}\right]$. Using the cumulative distribution function.

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30 [Intermediate] • Target: Comparing ellipses of the same eccentricity

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?

a. $8 - 2\sqrt{6}$ ✓

b. 4

c. $\frac{10}{3}$

d. $2\sqrt{6}$

1. **Find the total area of ΔABC:** The base along the x-axis connects $A(0, 0)$ and $B(8, 0)$, so the base length is $b = 8$. The vertex $C(2, 6)$ gives a height of $h = 6$. $\text{Area}(\Delta ABC) = \frac{1}{2} \times 8 \times 6 = 24$ Each of the two regions must therefore have an area of $\frac{24}{2} = 12$. 2. **Determine the location of $x = k$:** The vertical line through C is $x = 2$. The portion of ΔABC to the left of $x = 2$ is a triangle with base on the x-axis from $x = 0$ to.

31 [Intermediate] • Target: Standard form

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. **44** ✓

b. 43

c. 11

d. 22

We solve the problem in two stages: **Step 1: Determine σ** Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$. The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$: $\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$ $\frac{0.658}{\frac{\sigma}{5}} = 1.645 \Rightarrow \frac{3.29}{\sigma} = 1.645 \Rightarrow \sigma = \frac{3.29}{1.645} = 2.0$ mm **Step 2: Determine the minimum sample size m** For.

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32 [Intermediate] • Target: Applications

In the standard (x, y) coordinate plane, the vertices of ΔOAB are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides ΔOAB into two regions of equal area. What is the value of k?

a. $6 - 2\sqrt{3}$ ✓

b. $\sqrt{6}$

c. $\frac{8}{3}$

d. 3

1. **Find the total area of ΔOAB:** Taking the side along the x-axis as the base, the length of base OA is $6 - 0 = 6$. The height is the y-coordinate of vertex $B(2, 4)$, which is 4. $\text{Area}(\Delta OAB) = \frac{1}{2} \times 6 \times 4 = 12$ To divide the triangle into two regions of equal area, each region must have an area of $\frac{12}{2} = 6$. 2. **Determine the domain of k:** - If 02, the line $x = k$ lies to the right of vertex $B(2, 4)$. The region to the right of $x = k$ is a t.

33 [Intermediate] • Target: Ellipses centered away from the origin

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. **-9** ✓

b. -4

c. 4

d. 9

A system of two linear equations in two variables has no solution if and only if the lines represented by the equations are parallel and distinct (they have the same slope but different y-intercepts). First, rewrite each equation in slope-intercept form $y = mx + b$: 1. For the first equation:
$3x - 2y = 8 \Rightarrow -2y = -3x + 8 \Rightarrow y = \frac{3}{2}x - 4$ The slope of the first line is $m_1 = \frac{3}{2}$ and the y-intercept is -4. 2. For the second equation.

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34 [Intermediate] • Target: Eccentricity and orbital applications

If the real numbers x and y satisfy the system of equations $x^3 - 3xy^2, = 10; 3x^2y - y^3, = 5$ what is the value of $x^2 + y^2$?

a. $\sqrt{5}$

b. 5 ✓

c. $5\sqrt{5}$

d. 25

We can solve this system using algebraic expansion or complex numbers. **Method 1: Squaring and Adding** Square both equations and add them together: $(x^3 - 3xy^2)^2 + (3x^2y - y^3)^2 = 10^2 + 5^2 = 100 + 25 = 125$ Expanding the left-hand side: $(x^6 - 6x^4y^2 + 9x^2y^4) + (9x^4y^2 - 6x^2y^4 + y^6) = x^6 + 3x^4y^2 + 3x^2y^4 + y^6$ Notice that this expression is the binomial expansion of $(x^2 + y^2)^3$: $(x^2 + y^2)^3 = 125$ Taking the real cube root of both sides gives: $x^2 + y^2 = \sqrt[3]{125} = 5$ **Method 2.

35 [Intermediate] • Target: Graphing an ellipse

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

a. A = 1/6, D = 5/3

b. **A = 1/6, D = 10/3** ✓

c. A = 1/3, D = 5/3

d. A = 1/3, D = 10/3

We are given the velocity function $v(t) = \frac{1}{2}t^2 - 3t + 5$. Since displacement s is the integral of velocity with respect to time, we have: $s(t) = \int v(t)\,dt = \int \left(\frac{1}{2}t^2 - 3t + 5\right)dt$ Integrating term by term, we get: $s(t) = \frac{1}{2} \cdot \frac{t^3}{3} - 3 \cdot \frac{t^2}{2} + 5t + K\ s(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 5t + K$ Comparing this with the given form $s(t) = At^3 + Bt^2 + Ct + D$, we can identify $A = \frac{1}{6}, B = -\frac{3}{2}, C = 5$. The con.

36 [Intermediate] • Target: The ellipse's defining property

In the system of linear equations below, k is a real constant. $(k - 1)x + 6y, = 18; 2x + ky, = 12$ For which value of k does the system have infinitely many solutions (x, y)?

a. -4

b. -3

c. 3

d. 4 ✓

For a system of two linear equations in two variables to have infinitely many solutions, the two equations must represent the exact same line. This requires the ratios of the corresponding coefficients and the constant terms to all be equal: $\frac{k-1}{2} = \frac{6}{k} = \frac{18}{12}$ First, simplify the ratio of the constant terms: $\frac{18}{12} = \frac{3}{2}$ Next, set the ratio of the y-coefficients equal to $\frac{3}{2}$: $\frac{6}{k} = \frac{3}{2} \Rightarrow 3k = 12 \Rightarrow k = 4$ Verify this value.

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37 [Intermediate] • Target: Writing equations from a description

In the standard (x, y) coordinate plane, at what values of x do the graphs of $y = x^2 - 3x$ and $y = 2x - 4$ intersect?

a. $x = 1$ **and** $x = 4$ ✓

b. $x = -4$ and $x = -1$

c. $x = -2$ and $x = 2$

d. $x = 2$ and $x = 3$

To find the x-values of the intersection points, set the two expressions for y equal to each other: $x^2 - 3x = 2x - 4$ Subtract $2x$ and add 4 to both sides to write the quadratic equation in standard form: $x^2 - 5x + 4 = 0$ Factor the quadratic expression: $(x - 1)(x - 4) = 0$ Setting each factor equal to zero gives: $x - 1 = 0 \Rightarrow x = 1$ $x - 4 = 0 \Rightarrow x = 4$ Thus, the graphs intersect at $x = 1$ and $x = 4$. Misconception notes: - Choosing $x = -4$ and $x = -1$ typically results from sign.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Systems with ellipses • Cross-Lesson: 1.02 Domain and Range

In the standard (x, y) coordinate plane, what is the length of the major axis of the ellipse defined by the equation $\frac{(x-4)^2}{49} + \frac{(y+2)^2}{16} = 1$?

a. 7

b. 8

c. **14** ✓

d. 49

The standard form of a horizontal ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, where $a > b > 0$. 1. Identify the values of a^2 and b^2 from the given equation: $a^2 = 49$ and $b^2 = 16$ 2. Solve for the semi-major axis length a: $a = \sqrt{49} = 7$ 3. The total length of the major axis is $2a$: $2a = 2 \times 7 = 14$ Common misconceptions: - Choosing 7 confuses the semi-major axis length a with the full major axis length $2a$. - Choosing 8 calculates the length of the.

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39 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 1.03 Average Rate of Change

A circle has its center at the point $(2, -3)$ and a radius of 5 units. Which of the following is the standard form equation of this circle?

a. $(x-2)^2+(y+3)^2=25$ ✓

b. $(x+2)^2+(y-3)^2=5$

c. $(x-2)^2+(y+3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

The standard form equation of a circle with center (h, k) and radius r is given by $(x-h)^2+(y-k)^2=r^2$. In this problem, the center is $(h, k)=(2, -3)$ and the radius is $r=5$. Substituting these values into the standard form equation, we get $(x-2)^2+(y-(-3))^2=5^2$, which simplifies to $(x-2)^2+(y+3)^2=25$.

40 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Derive the reflection property: light from one focus of an ellipse reflects to the other.

a. **By Snell's law variant: angle of incidence = angle of reflection using the definition $PF_1+PF_2=2a$ and tangent bisects angle F_1PF_2** ✓

b. Only for circles

c. Only for parabolas

d. Arbitrary property

Applying Conic Sections principles, Optical property of ellipse. Therefore, the verified correct option is By Snell's law variant: angle of incidence = angle of reflection using the definition $PF_1+PF_2=2a$ and tangent bisects angle F_1PF_2.

41 [Challenging] • Target: Standard form • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the polar equation of an ellipse with focus at the origin, semi-major $a=5$, $e=0.6$.

a. $r=\frac{5}{1-\cos\theta}$

b. $r=\frac{5}{1+0.6\cos\theta}$

c. $r=\frac{3.2}{1-0.6\cos\theta}$ ✓

d. $r=\frac{8}{1+0.6\cos\theta}$

For an ellipse with focus at the pole, $r=\frac{a(1-e^2)}{1-e\cos\theta}=\frac{5(1-0.36)}{1-0.6\cos\theta}=\frac{3.2}{1-0.6\cos\theta}$.

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42 [Challenging] • Target: Applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Convert the hyperbola $4x^2 - 9y^2 = 36$ to standard form.

a. $\frac{x^2}{9} - \frac{y^2}{4} = 1$ ✓

b. $x^2 - y^2 = 9$

c. $\frac{x^2}{36} - \frac{y^2}{36} = 1$

d. $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Applying Conic Sections principles, Divide by 36: $x^2/9 - y^2/4 = 1$. Therefore, the verified correct option is $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

43 [Challenging] • Target: Ellipses centered away from the origin • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Convert to standard form: $(3 + 4i)(1 - 2i)$.

a. $3 - 8i$

b. $11 - 2i$ ✓

c. $11 + 2i$

d. $7 + 2i$

Applying Quadratics & Complex Numbers principles, $3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i$. Therefore, the verified correct option is $11 - 2i$.

44 [Challenging] • Target: Eccentricity and orbital applications • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the parametric equations for the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

a. $x = 4\sin t, y = 3\cos t$

b. $x = 4t, y = 3t$

c. $x = 4\cos t, y = 3\sin t$ ✓

d. $x = \cos t, y = \sin t$

Applying Conic Sections principles, Standard parametrization. Therefore, the verified correct option is $x = 4\cos t, y = 3\sin t$.

45 [Challenging] • Target: Graphing an ellipse • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A system of 3 equations in 3 unknowns is represented by $AX = B$ where A is:

a. 3×3 ✓

b. $3 \times B$

c. 3×1

d. 1×3

Applying Matrices & Systems principles, Coefficient matrix is 3×3. Therefore, the verified correct option is 3×3.

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